[PLACEHOLDER: the title claim is still to be written]

[PLACEHOLDER: Title TBD]

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The motivation pitch will connect housing access to fertility

[PLACEHOLDER: pitch]

tion fact 1

tion fact 2

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The paper contribution statement is still a placeholder

[PLACEHOLDER: this paper]

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The literature positioning will be filled after author review

rugge slot

rmann slot

(DUE) slot

acamo slot

A two-period OLG links young fertility choices to old incumbent retention

Potential young households enter with income and wealth:

$$i = (y_i, a_i),$$

so entry wealth directly shapes access to owner housing.

- Young households choose entry, tenure, occupied housing, fertility, and saving.
- Young owners become old incumbents; old renters enter old age without housing.
- Old incumbents choose consumption, retained housing, and estates.

The tax wedge from sale rather than death is summarized by:

$$\bar{\ell} = \tau^g \, \frac{P\,\varpi}{1+\varpi}$$

Selling triggers the capital-gains tax, while death forgives it through basis step-up.

Rental and finance frictions collapse into one family-housing cap

The owner cap is the smaller of the owner size limit and down-payment capacity:

$$h_i \leq H_i^O(q, \tau^p) = \min\left\{\bar{h}^O, \frac{a_i \rho}{(1-\phi)q}\right\}.$$

The renter analog is $H_i^R = \bar{h}^R$, so tenure maps different primitives into one cap object. The child-cost display makes the cap's fertility price exact:

$$\underbrace{\frac{\vartheta\left(c_i^m - \chi n_i^m\right)}{n_i^m}}_{\text{willingness to pay for a marginal child}} = \underbrace{\chi + \kappa(q^m + \zeta_i^m)}_{\text{full price of a child}}.$$

A binding access constraint acts like a tax of $\kappa \zeta_i^m$ per child.

The access gap is the goods value of relaxing constrained family space

For renters, the bundle reveals the cap bite:

$$\frac{u^Y_{h,i}}{u^Y_{c,i}} = \frac{\alpha\,(c^R_i - \chi n^R_i)}{h^R_i - \kappa n^R_i} = q + \zeta^R_i.$$

A renter at the size cap values another unit above market rent.
For owners, the access gap separates finance from the owner size limit:

$$\zeta^O_i = \underbrace{\frac{\mu_i}{\lambda^O_i}(1-\phi)P}_{\zeta^{O,F}_i,\ \text{down-payment component}} + \underbrace{\frac{\theta^O_i}{\lambda^O_i}}_{\text{owner-size-limit component}} \geq 0.$$

The gap is computable from consumption and housing bundles, net of child needs.

Where the cap binds is where fertility responds

Under lifetime equal scaling, the fertility derivative is proportional to the same gap that measures access:

$$\Psi_i^m = \frac{\kappa}{\alpha}\frac{\zeta_i^m}{c_i^m - \chi n_i^m}\Omega_i^m, \qquad \Omega_i^m = \frac{\frac{\alpha}{h_i^m - \kappa n_i^m} + \frac{q^m}{c_i^m - \chi n_i^m}}{\frac{\vartheta}{(n_i^m)^2} + \frac{\chi^2}{(1+k)(c_i^m - \chi n_i^m)^2} + \frac{\alpha\kappa^2}{(h_i^m - \kappa n_i^m)^2}} > 0.$$

Holding exposure and pass-through fixed, larger normalized gaps predict larger birth responses.

Takeaway. Where the cap binds is where fertility responds.

The equilibrium misallocates stock when constrained young face locked-in incumbents

The planner comparison moves one unit from an old incumbent to a financially constrained young owner:

$$\underbrace{\left(q^O + \zeta_i^{O,F}\right)}_{\text{buyer's marginal value}} \quad - \quad \underbrace{(q - \bar{\ell})}_{\text{incumbent's marginal value}}$$

$$= \zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell} > 0 \quad \text{for an improving reallocation.}$$

The market price cancels; the remaining surplus is financing access plus the two tax wedges.

Takeaway. The misallocation is not old occupancy per se, but old lock-in coexisting with young access gaps.

The quantitative model turns the OLG mechanism into a lifecycle housing ladder

- Lifecycle: **[NUMBER NOT FOUND: J=17 four-year periods and ages 18–82 in the named source files]**.
- Income and children: **[NUMBER NOT FOUND: 5-state income process and parity 0–2 with child stages in the named source files]**.
- Tenure: renting is capped at 6 rooms; the owner ladder is **[NUMBER NOT FOUND: [2,4,6,8,10] owner-room ladder in the named source files]**.
- Financing and market: the down payment is $(1-\phi)$ with $\phi = 0.80$; one housing market determines q in general equilibrium.

The maintained link to Part I is that the down-payment term enters $\zeta_i^{O,F}$ through $(1-\phi)P$.

The calibration is overidentified in moments, but some setup counts need source confirmation

- The reported fit is a 14-moment SMM table; the free-parameter count is **[NUMBER NOT FOUND: 13 free parameters in the named source files]**.
- The search is performed at $N_b = 120$ and final verification is at $N_b = 240$.
- The empirical inputs include ACS room moments; **[NUMBER NOT FOUND: PSID use in the named source files]**.
- The rental cap is not estimated: $\bar{h}^R = 6$ rooms is imposed externally by DUE's p90 rental-size rule in the authors' ACS sample.

The cap note's precise statement is that the p90 falls in the 6-room bin, not that p90 equals 6.000.

The rental cap is identified by a sharp loss valley near the external rule

The cap profile compares the imposed DUE-rule value to nearby alternatives:

$\bar{h}^R$	best loss	state
5.5	7.69	converged
6.0	6.31	converged
6.5	6.24	still improving
7.0	7.15	wave A only

The data mildly prefer 6.5, but both flanks rise relative to the external value.

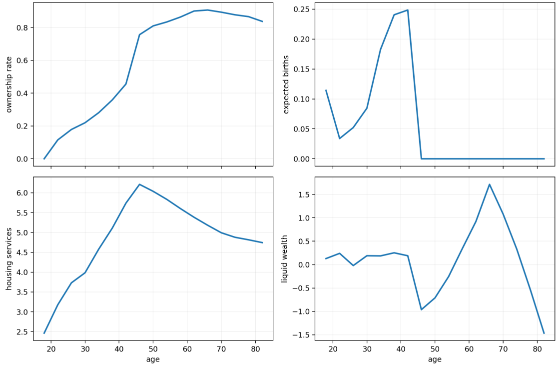
Takeaway. There is a sharp valley near where the external DUE rule places the cap, so the data do not fight the imposed value.

The honest Nb=120 calibration fits paper-core moments but leaves three misses

moment	target	model	gap	contrib
tfr	1.918	1.947	+0.029	0.02
childless_rate	0.188	0.260	+0.072	0.10
own_rate	0.576	0.531	-0.045	0.20
own_family_gap	0.168	0.298	+0.131	0.77
housing_increment_0to1	0.664	0.617	-0.048	0.03
old parent-childless wealth gap 65–75	1.007	0.835	-0.172	0.06
childless renter mean rooms	3.805	3.656	-0.150	0.13
childless owner share rooms $\geq$ 6	0.596	0.595	-0.001	0.00
old nonhousing wealth/income median	2.231	2.328	+0.097	0.01
young renter wealth/income	0.179	0.354	+0.175	0.37
owner–renter mean rooms	**2.419**	**2.152**	**-0.266**	**0.85**
old_age_own_rate	**0.764**	**0.892**	**+0.128**	**2.63**
own_rate_2534	**0.341**	**0.226**	**-0.115**	**1.06**
parent 3+ vs 1–2 rooms gap	0.368	0.264	-0.104	0.09

Takeaway. Paper-core moments are close; old-age ownership, young ownership/wealth timing, and childlessness remain diagnosed misses.

The lifecycle profiles show a deadline purchase and delayed fertility



- Ownership jumps at the last fertile age.
- Births occur at 33 with a deadline spike; the data benchmark is ~ 27.
- Liquid wealth falls sharply at the family purchase, consistent with maximum leverage.

Takeaway. The fit problem is timing: the model buys and gives birth too late.

The remaining misses point to missing mechanisms rather than a new local search

diagnosed miss	designed but not adopted fix
Old-age ownership is 0.892 in the model vs 0.764 in the data.	Add a health, mortality, or downsizing exit margin for old owners.
Young ownership is 0.226 vs 0.341, and young wealth is 0.354 vs 0.179.	Address fertility timing rather than only re-searching the same box.
Childlessness by income goes the wrong way relative to the data.	Add a child time cost so children are not priced only in goods and rooms.

Takeaway. The miss pattern is coherent: late fertility and absent old-age exit do work that parameters cannot do cleanly.

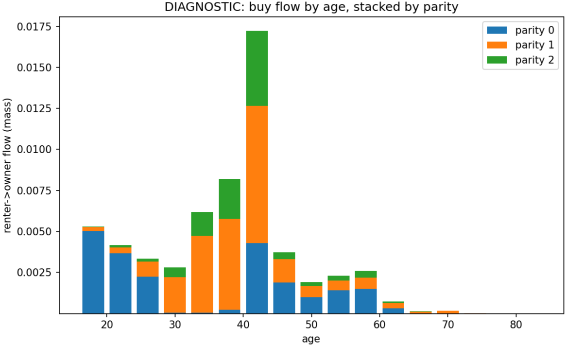
The measured access gap is the theory prediction showing up quantitatively

- Among cap-pinned fertile renters, $\zeta/q = 1.28$.
- These renters value family space at $2.3\times$ market user cost.
- 0.734 of fertility-derivative mass sits on $\zeta > 0$ households.
- The result is grid-converged, robust across re-weighted calibrations, and ladder-invariant.

The equal-scaling result says that Ψ_i^m rises with the normalized gap $\zeta_i^m/(c_i^m - \chi n_i^m)$.

Takeaway. This is the Ψ-proportional-to-ζ theory prediction in the calibrated model.

Ownership is triggered by births at the deadline



- **[NUMBER NOT FOUND: 29 percent of the age-42 cohort buys in one period in the named source files]**.
- **[NUMBER NOT FOUND: 65 percent of parent buyers purchase within the birth period in the named source files]**.
- **[NUMBER NOT FOUND: buyers arrive roughly 10 grid cells above the down-payment threshold in the named source files]**.

Takeaway. The purchase margin is a birth-timing margin, not slow saving alone.

Family-sized housing is mostly held outside the fertile-parent margin

The calibrated allocation gives the empirical face of the misallocation proposition:

holder group	share of rooms ≥ 6 stock
post-fertile households	87%
households with no kids at home	72%
fertile parents	12.6%

The households with the largest measured access gap hold a small share of family-sized owner stock.

Takeaway. This is the misallocation proposition's empirical face.

A price instrument capitalizes into large-home prices but barely moves births

- The policy is a 2% tax on large homes.
- The large-home price effect is **[NUMBER NOT FOUND: -12.3 percent large-home price response in the named source files]**.
- The tax-alone fertility response is $\Delta\mathrm{TFR} = +0.004$.
- Robustness is reported as grid, menu, and weight invariant in the mechanism notes; the named bridge file pins the birth effect.

Takeaway. This instrument capitalizes into prices; it does not create births.

Birth-linked purchase grants move fertility through access

The grant dose-response is monotone in the access amount:

grant size	$\Delta\mathrm{TFR}$
0.1	$+0.01$
0.4	$+0.13$
0.58	$+0.26$
1.0	$+0.49$

Prices move **[NUMBER NOT FOUND: roughly +1 percent grant price response in the named source files]**.

- Caveat: small conditional grants can delay pre-birth purchases through anticipation effects.

Takeaway. Access grants move the birth margin because they relax $\zeta_i^{O,F}$.

The policy package decentralizes the planner's reallocation

The package taxes the incumbent stock and uses the revenue to fund young buyers' access:

$$\text{large-home tax} + \text{birth-linked purchase grant} \quad \Rightarrow \quad \Delta\mathrm{TFR} = +0.15, \quad \Delta p_{\text{large}} = -11\%.$$

The bridge note reports this package as revenue-funded.

Takeaway. Access, not price, is the fertility margin.

The bridge is access gaps, stock misallocation, and decentralized reallocation

1. Measured access: capped fertile renters have $\zeta/q = 1.28$, and 0.734 of the derivative mass sits on $\zeta > 0$.
2. Stock allocation: 87% of family-sized stock is held by post-fertile households.
3. Policy: the tax-funded access transfer raises TFR by 0.15 while cutting large-home prices by 11%.

The bridge note's synthesis is that the same ζ object measures fertility exposure and misallocation, and the package is the decentralized version of the planner reallocation.

The next version needs mechanisms for bounds, timing, and childlessness

- **[NUMBER NOT FOUND: 8 of 13 calibrated parameters at search bounds in the named source files]**.
- The fertility-timing issue remains: births occur at 33 while the data benchmark is ~ 27.
- Designed extension: add a child time cost to price children partly in foregone income.
- Designed extensions: add an old-age exit margin and estimand-consistent childlessness targets.

Takeaway. The draft should present these as limitations, not as solved calibration details.

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